FedScreen — Future Work: Detecting More Than One Disease

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Where we are today

FedScreen checks one disease per image, and it does that well.

- One shared ConvNeXt-Tiny backbone feeds three small heads: malaria, anemia and leukemia.
- A specimen router recognises the image type and runs only the matching head.
- The other two diseases are shown as negative, because they were not tested on that image.

Why it cannot say "two diseases" yet

The model has never seen a patient with more than one condition.

- Each dataset is a different specimen (malaria cell, red-cell field, white-cell crop), and every image has exactly one label.
- Each head only learned its own cell type, so it cannot judge the others.
- The router picks one head, so by design only one answer comes out.

Roadmap to detect 2 or 3 diseases together

1. **Patient-level report (no new data needed).** One patient has many images. Run the router and the matching head on each image, then combine the results into one report, for example "malaria positive, anemia positive".
2. **Multi-label training.** Replace the two-way softmax with one independent yes/no (sigmoid) output per disease, trained with binary cross-entropy. Then several outputs can be positive at once.
3. **Slide-level model.** Let the network look at many cells from one slide and weigh the important ones (attention pooling), so one patient gets one set of disease probabilities.
4. **Use disease links.** Some diseases often occur together, such as malaria and anemia. Calibrate each probability and tune per-disease cut-offs so the combined output is trustworthy.

What we need and how we will measure it

- **Data:** patients with confirmed multiple diagnoses and several smear images each, from partner hospitals, with ethics approval. Mixing images from different people would not be valid evidence.
- **Splits:** keep every patient entirely in train, validation or test.
- **Metrics:** per-disease AUC and F1, plus how often the full set of diseases is exactly right, and sensitivity for each combination (for example malaria with anemia).
- **Federated setting:** keep FedProx, since each health centre will see a different mix of conditions.

Safety note

FedScreen is a screening aid: a doctor confirms every finding, and the model should say "not sure" when it is uncertain.